VR Pseudo-Haptic mass emulation

Introduction

When a purely virtual object is manipulated, biological receptors in muscles and tendons are not activated and so other techniques are needed to give the brain the perception of weight (Samad, Gatti, Hermes, Benko, & Parise, 2019). Pseudo-haptics are a range of techniques that create the illusion of feedback, in this case object weight in VR. The Control/Display ratio (C/D ratio) refers to a specific technique where the VR visuals are offset from real world hand movements. This creates the illusion of weight by moving virtual hands at a slower rate than real world user hand movements.

The scope of this technique in this paper is limited to vertical object lifting and focussed on a single hand. While this is the articulated scope, this technique could be trivially applied to horizontal movements such as pushing a heavy door and is generally applicable more broadly.

This paper will outline the technique and implementation details followed by a deeper dive into the literature basis for the technique itself. The paper will round out by discussing a range of potential suitable ‘interaction vault’ categories applicable and discuss future work to refine and improve the technique.

Technique Description

The pseudo-haptic weight emulation for VR technique involves visually offsetting the visuals of a lifting object by an amount related to the force required to move that object. As a practical example, where virtual hands are rendered, when a user lifts a heavy object, if their hands physically move up 30cm, the rendered hands (holding the object) may only move up 15cm. When the user lifts a lighter object, the distance between real and rendered hands may only be 2cm (i.e.. the hands move up 28cm). This visual offset provides the cognitive stimulation of object weight. The research supporting this and detailed considerations are included in the following section with the formula derivation discussion located in the Mathematical discussion subsection.

To implement this weight emulation, the following steps are taken by the object weight emulation component:

- Initial force application detection
- C/D ratio offset
- Force release

Preliminaries

Before implementing this component, design decisions need to be made (as discussed in the Mathematical discussion subsection). The primary design consideration is ensuring the C/D ratio for objects do not prevent effective movement. If users need to be able to lift an object in the experience to 80% of their max reach, it is important to ensure the C/D ratio for that weight is at least 0.8. Failure to consider this may result in objects not being able to be physically moved to the intended location within the user's reach. The primary mechanism for this design is the *Reference Weight* (W_{ref}). This is the weight used for visual offset baselines and should generally be twice the mean object weight in the experience. If there is a wide range of weights, the maximum weight may need to be used for this value instead. This value practically represents the weight at which 50% of movement is translated into the virtual world, ie. a real-world hand movement of 10cm vertically would translate into a 5cm virtual vertical movement.

The scaling factor SF is the core aspect of this component which determines the visual movement rate offset and is calculated with the below formula where k is a tuning factor and m is the mass of the virtual object. Note that k in the formula is a tuning factor; it is strictly positive and defaults to 1 but can be modified for specific feel.

$$SF = \frac{1}{1 + k\frac{m}{W_{ref}}}$$

The scaling factor formula represents the proportion of real-world movement that is represented in the virtual environment. Figure 1 shows this factor calculation where the mean object weight is 5kg or the max weight is 10kg, i.e. $W_{ref} = 10$.

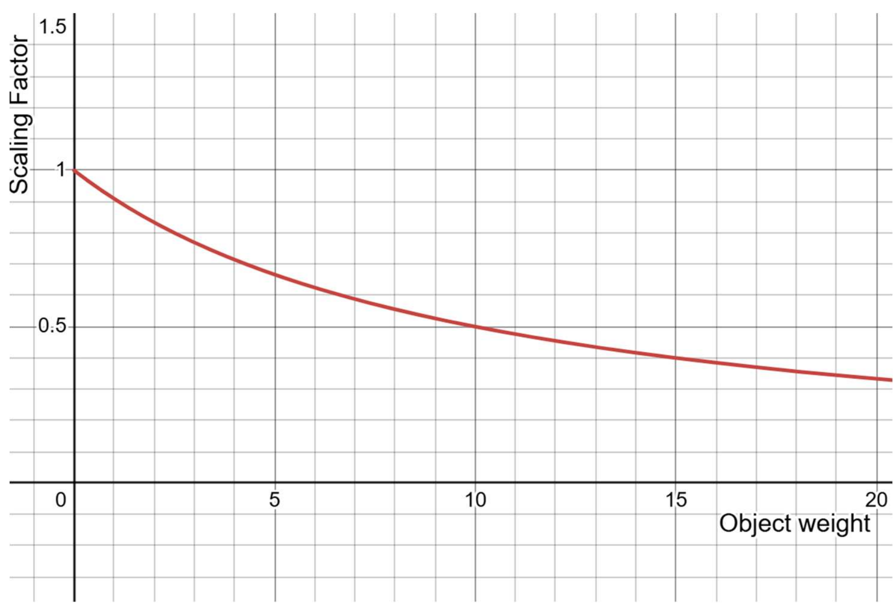

Figure 1 - Scaling factor calculations for an experience with objects having a mean weight of 5kg.

Initial force application detection

The first step is the object lift detection. This technique is implementation agnostic, but the triggering of the lift commences the C/D ratio offset updates and stores the object grabbed location.

C/D ratio offset updates

The ongoing C/D ratio updates involve scaling the movement in virtual space based on the calculated Scaling Factor. Note the formula provided here is a start point and can be tailored to individual experiences.

The simple implementation of these updates is to track relative movement frame by frame and scale the vertical aspect by the Scaling Factor. A more advanced approach that is less susceptible to floating point issues is to store the original pickup location and scale the vertical coordinate based on the total vertical delta from that reference point each frame.

As an example:

- Frame 1
    - Lift point: (10, 4, 10) – where (x, y, z) vector has y as the "up" direction.
    - Scaling factor: 0.9
- Frame 2 hand location: (11, 5, 10):
    - Delta between locations: (1, 1, 0).
    - Vertical delta scaled by scaling factor: $1 \times 0.9 = 0.9$
    - Scaled delta: (1, 0.9, 0)
    - Visual location = Initial lift point + Scaled delta
    - Visual location frame 2 = (11, 4.9, 10)

This underlying approach is the same for both approaches; the only difference being that "Frame 1" in the simple implementation is the previous frame, whereas in the latter implementation, "Frame 1" is the frame where the initial lift began.

Force release

The final element of this technique is the C/D ratio removal on release. At the release stage the visual offset needs to be undone. This should be done via a rapid interpolation between visualised position and real hand position.

Additional considerations

While out of scope of this paper, there are several aspects to consider:

- This technique is focussed on the lifting phase under a single hand paradigm. Additional complexities are introduced where a second lifting point is used but in the simple case of vertical lift, halving the item mass is a suitable approximation. Note this will have edge cases where hands are able to grip/ungrip that will need

to be addressed including rapid C/D ratio changes and real-world hand vertical offset desynchronisation.
- This technique focusses on the lifting phase; research discussed in the Cognitive impact thresholds subsection offers detail relating to relative sensitivities in all spatial dimensions. This information will inform dimensional scaling in any broader application. Specifically, Figure 6 provides thresholds where user detection is likely by spatial axis; note that this is not necessarily a negative within the context of the experience but can be used to guide Scaling Factor tuning, in particular across relative spatial axis.
- This technique has solely focussed on the C/D ratio. Studies have shown that a "force vector" visualised can aid in the weight emulation; if this is an appropriate visualisation for the use case it is worth considering including this aspect.
- Additional pseudo-haptics (audio, VFX etc) are also able to be implemented and scaled based off the inverse C/D ratio (the lower the C/D ratio, the more "strain" is caused).

Technique Discussion

Theoretical base

Using C/D ratio for object weight emulation has been studied at some length with demonstrable outcomes. A study by Dominjon et al (2005) involved participants moving a ball hooked up to force feedback up and down out of participant vision. The participant observed a visualisation of the ball on a monitor and responded to survey questions. In this case it was identified that C/D ratio strongly modified the perception of mass where a value of greater than one indicated a higher weight (Dominjon, Lecuyer, Burkhardt, Richard, & Richir, 2005).

Samad et al (2019) extended this through a study in which real objects being manipulated were rendered in VR. The study approach used optically tracked objects with a C/D ratio of 1.25 and 0.75 and users observing VR representations of the object's movement, as per Figure 2. In this study the initial activity was fully freeform with researchers noting participant reactions. In this phase 50% of users spontaneously mentioned a feeling of weight perception and in the subsequent phase where researchers posed queries, 100% of participants reported higher C/D objects seeming heavier when directly questioned (Samad, Gatti, Hermes, Benko, & Parise, 2019). This former point in particular is a strong indicator that this technique may be suitable for VR weight emulation.

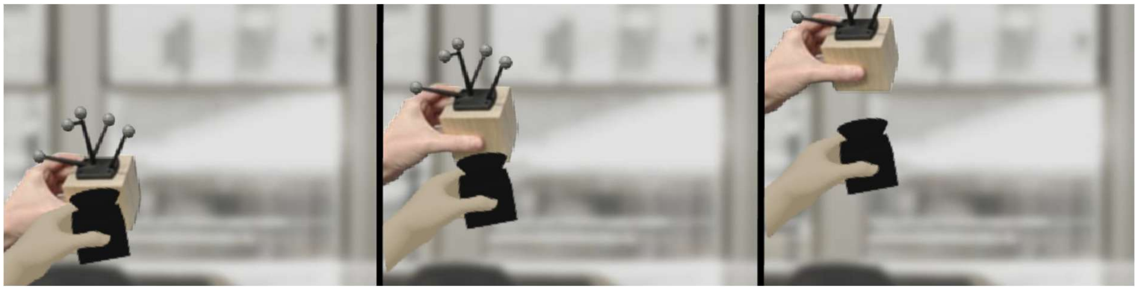

Figure 2 - Control/Display Ratio manipulation visualised (Samad, Gatti, Hermes, Benko, & Parise, 2019)

While the studies observed so far provide a clear indication that pseudo-haptics (specifically through C/D ratios) give the user an indication of weight, Palmerius et al (2014) investigated the difference between haptics and pseudo-haptics. In particular their study considered a 'cross modal effect' between haptics and pseudo-haptics. The study used a 'force causes displacement' approach and involved multiple cubes with varying haptic and pseudo-haptic feedback levels with the goal being to isolate the relative impact of both on weight perception; the physical and VR setup of this experiment is included as Figure 3.

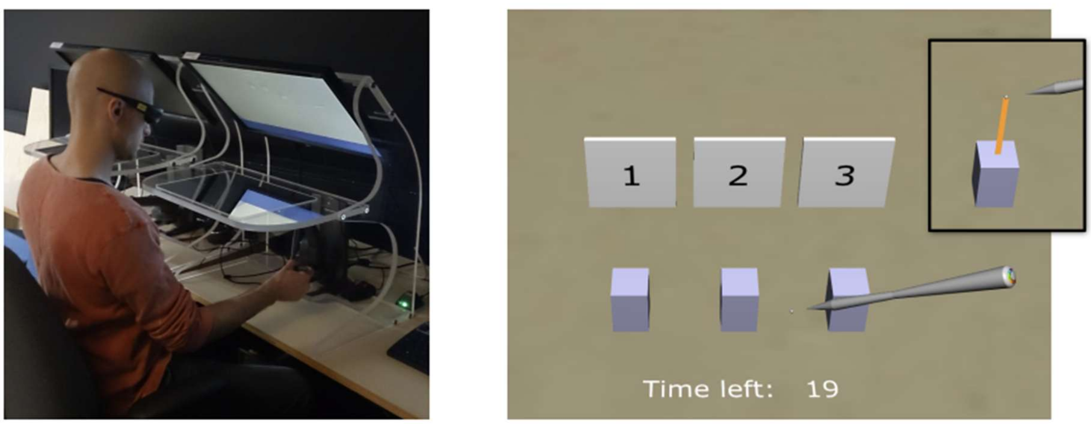

Figure 3 - Force causes displacement study setup (Palmerius, Johansson, Höst, & Schönborn, 2014)

Palmerus et al (2014) demonstrated the impact of C/D noting that '*participants were significantly more successful in the condition with pure visual weight feedback than in any of the other three conditions*', including the pure haptic feedback condition. One perhaps counter-intuitive aspect of this study was the multi-modal feedback of both haptics and pseudo-haptics was less effective than the purely visual haptics although the authors note that '*at the conscious level at least, the received bi-modal sensory input was judged to invoke a degree of intermodal sensory conflict*' (Palmerius, Johansson, Höst, & Schönborn, 2014).

The psychological mechanism for weight perception was demonstrated in a more recent study which in particular noted that "*similar sensations of fatigue and changes in hand kinematics during both virtual object tasks, resulting from pseudo-haptic* *feedback, and real weight lifting tasks*" (Moosavi, Raimbaud, Guillet, & Mérienne, 2024). This study involved participants in VR moving a bottle placed on a box, vertically to a goal area, marked by a 'virtual window'; this VR view is shown in Figure 4. This study both used real objects (a real bottle placed on a real-world box with VR setup aligned to real world) and fully virtual objects; the latter including both VR reference to baseline the impact of VR controller weight, and pseudo-haptic measurements.

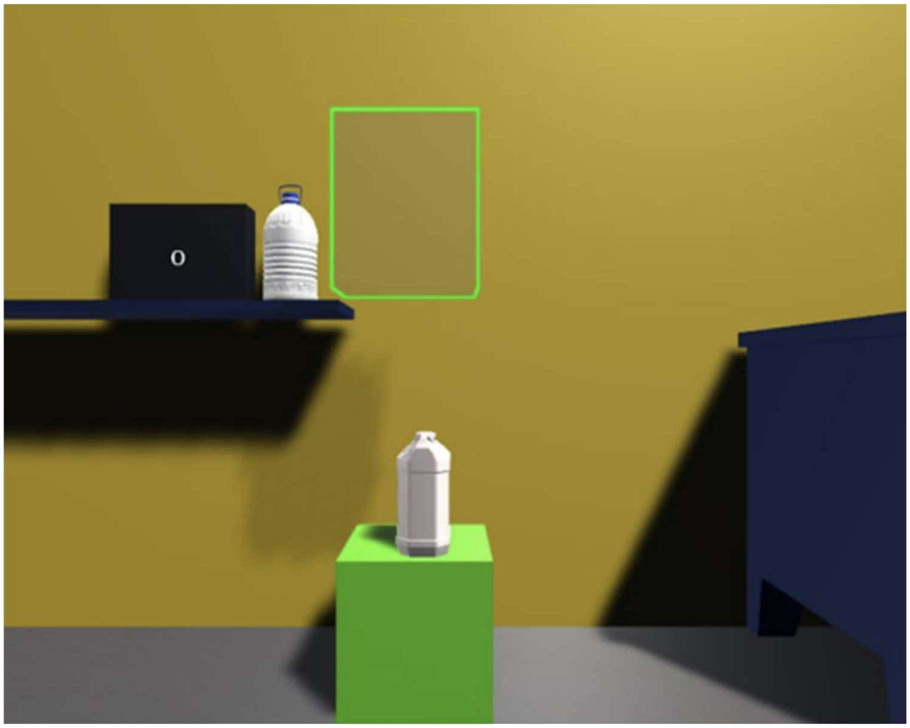

Figure 4 - Experimental VR setup for 'Enhancing weight perception in virtual reality: an analysis of kinematic features' (Moosavi, Raimbaud, Guillet, & Mérienne, 2024)

This discovery is a critical indicator that pseudo-haptic weight emulation is more than just a visual indicator and importantly, Moosavi et al (2024) demonstrated this effect clearly in a pure VR sense, with no real-world objects present.

Refined Discussion

Limitations of note

It should be noted that the research used in the analysis of this technique focussed on a controlled environment, in particular using single hand movements. As Samad et al (2019) note in their approach, *"they could not use two hands to grasp a single cube because of the offset that would have been accumulated for the hand that had picked it up from the table, and the lack thereof for the newly grasping hand, which would lead to a differential visual-proprioceptive incongruence across the hands"*.

It is also worth noting that in another study, the movement duration for pseudo-haptic weight emulation was notably higher than both the real world and VR reference (Moosavi, Raimbaud, Guillet, & Mérienne, 2024). This is not a limitation but a design consideration to bear in mind; interactions are likely to be slower than real world so design will need accommodate for this.

While the study by Palmerius et al (2014) noted that the multi modal condition did not have any impact on the perception of weight, they also note that drawing a line between the displayed object and point of attachment has *'been shown to make users explicitly associate the visual impression with the forces involved in the interaction* (Palmerius, Johansson, Höst, & Schönborn, 2014).

Cognitive impact thresholds

One concern of this technique may be the difference between the real-world body position and the visualised position resulting in cognitive challenges. This has not been mentioned in the previously discussed studies and cognitive exercises such as the Rubber Hand Illusion appear to confirm that visual offsets are not a cognitive blocker when embodiment is sufficient. Burns, et al (2005) investigated the limits of a user's ability to detect positional offset between their virtual and real hand with the main finding being users are generally unaware of offsets until they are greater than 3-4.5cm. It was also noted that there were no specific differences in vertical vs horizontal offsets although it was acknowledged that this may have been a study limitation (Burns, et al., 2005).

Hartfill, et al (2021) investigated in detail the detection thresholds for users with varying gain in virtual hand movement compared to user's real movement. The experimental approach is demonstrated in Figure 5 and importantly, considered perception as it related to user centred axis; forward, backwards, upwards, downwards, inwards and outwards.

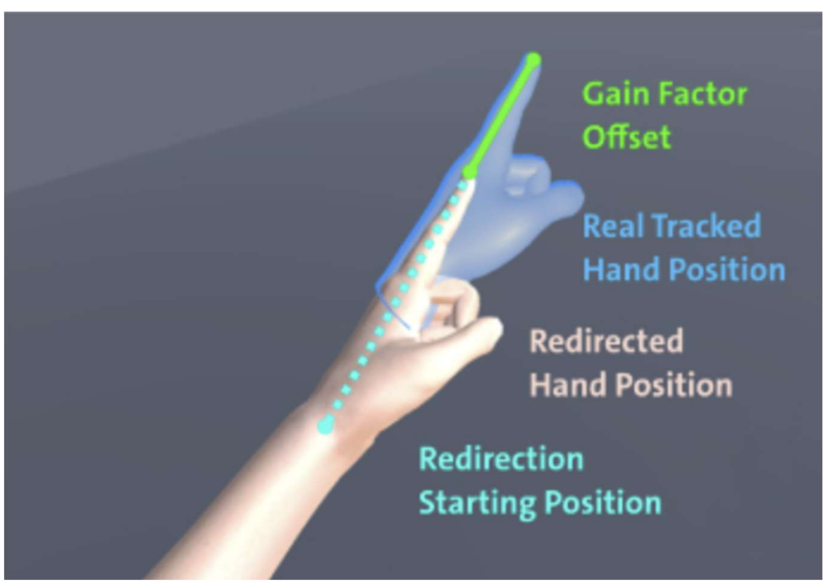

Figure 5 - Hand redirection experimental approach (Hartfill, et al., 2021)

Of note, this study identified significant differences between detection thresholds by spatial axis (Hartfill, et al., 2021), in particular upward and forward movements have a much higher detection threshold, i.e.. the ratio of real to virtual movement needs to be closer to 1.0 than other axes. Figure 6 contains the summary for all spatial directions in this study. This has implications for the weight emulation using pseudo-haptics; lifting an object vs pulling one down, pushing vs pulling an object will all have different weight perceptions if a static C/D ratio is used.

Movement Direction	Hand	*M*	*SD*	DT
inward	dominant	.603	.137	**.618**
	non-dominant	.633	.167	
outward	dominant	.566	.134	**.552**
	non-dominant	.538	.139	
upward	dominant	.711	.149	**.690**
	non-dominant	.669	.120	
downward	dominant	.598	.130	**.578**
	non-dominant	.558	.119	
forward	dominant	.748	.131	**.727**
	non-dominant	.705	.124	
backward	dominant	.592	.132	**.597**
	non-dominant	.603	.127	

Figure 6 - Mean (M) and standard deviations (SD) for Detection Thresholds for mid-air hand redirection with proposed Detection Threshold (DT) values (Hartfill, et al., 2021)

A fascinating psychology paper by Gibson (1933) offers some insight into the brain processes in context. The paper outlines experiments were participants observed curved lines for long periods of time; 'sensory adaption' from the brain resulted in the curves seeming less pronounced and subsequent straight lines appearing curved in the opposite direction. This suggests that the brain adapts to visual offsets over time and implies that the removal of these offsets can have a negative counter impact; in this case we may expect to see objects appearing lighter over time. This is supported by pseudo-haptic specific research. A recent study demonstrated that as the range of repetitions of lifting a virtual weight increased from 10 to 100, an object with a C/D ratio of 0.8 perceived weight approached the C/D ratio of 1.0 (Ito, Ban, & Warisawa, 2024); the results of which are included as Figure 7. This 'habituation' phenomenon supports the cognitive conclusions we draw from Gibson (1933).

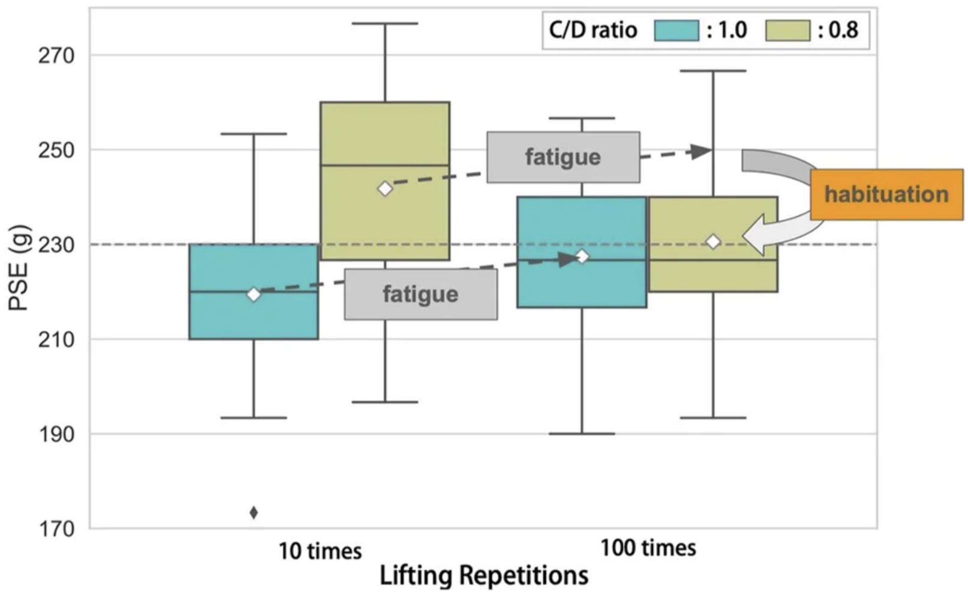

Figure 7 - Diminished effect of pseudo-haptic weight emulation on perceived weight of a virtual cube (PSE) (Ito, Ban, & Warisawa, 2024)

Mathematical discussion

A range of approaches were observed in the literature however these were largely focussed on determining weight. Samad et al (2019) numerically integrated instantaneous velocity and scaled by the desired C/D ratio to determine a location based offset. This simplifies into the graph shown as Figure 8.

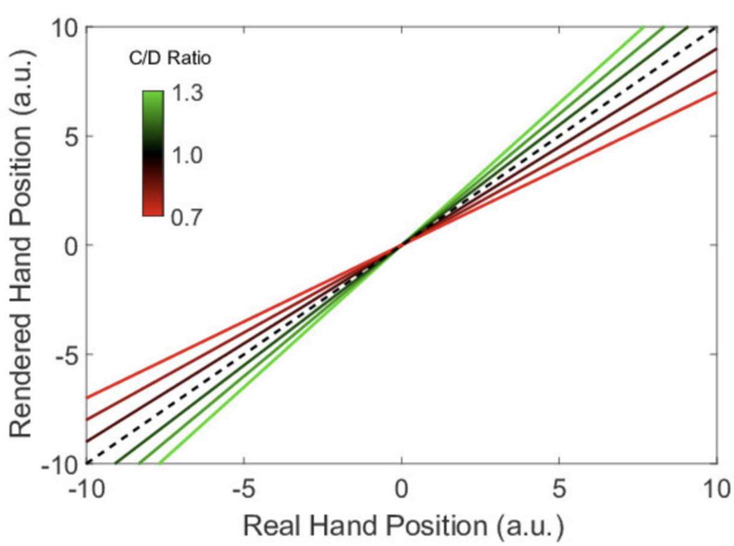

Figure 8 - Manipulation of C/D ratio for pseudo-haptic weight (a.u.: arbitrary units) (Samad, Gatti, Hermes, Benko, & Parise, 2019)

Samad et al (2019) also identified quantifiable predictions of weight perception based off the differing lift height, shown in Figure 9 however it is worth noting that the relative weight range is in the tens of grams.

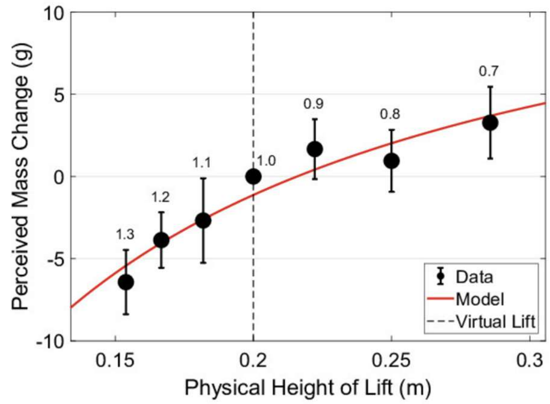

Figure 10: Perceived mass of the cube in grams as a function of the height of the lift performed in physical space. The vertical dashed line represents the height reached by the virtual hand (20 cm) and is added for clarity and to emphasize that the x-axis represents the height reached by the physical hand, and that data points away from the vertical dashed line represent greater discrepancies between the real and rendered movements. The numbers above each data point represent the C/D ratio corresponding to that height.

Figure 9 - Perceived weight of cube based off physical lift heights and C/D ratios (Samad, Gatti, Hermes, Benko, & Parise, 2019)

The pseudo-haptics baseline used by Palmerius et al (2014) consisted of an offset of 40mm per Newton of force. This approach has a more physics-based grounding and it is proposed that this is more suited to VR experiences where objects need a ‘weight’. The downside of this approach however is it is not fully scalable; if force requirements get too high, the offset will rapidly become too large to be feasible. As a result, the best approach is a function that approaches a maximal offset value asymptotically.

To effectively develop this, we need to consider both the maximum reach height of users, the maximum and mean weight of objects to be lifted and the maximum lift height required within the experience. With these details we are able to calculate the mass offset scaling factor which will maximise the ability to leverage visual offsets within the experience. Note this Scaling Factor will be the factor that the vector is scaled to so a heavy object will be < 1.0; this corresponds to a C/D ratio of >1.0. This is highlighted here to ensure terminology confusion is avoided.

Scaling factors can be used in a few different approaches:

- **Fixed offset.** The user needs to move their hand while lifting at least the offset before the object moves 1:1 with the user’s movements. This is simple to implement but not based in literature and will likely just result in a ‘sticky’ feeling.
- **Frame by Frame delta.** Each frame, scale the movement during the lifting phase by a scaling factor in the vertical direction. This is simple to implement but has the risk of accumulating floating-point imprecision.
- **Total movement scaling.** This achieves the same end state but records the initial lifting position and each frame identifies the vertical position based off the current hand position and the original lifting position. This vector is scaled by the

Scaling Factor to determine the current visualised position. This is slightly more complex and less flexible) but negates the cumulative error issue.

Technique Categorisation

The locomotion vault[1] provides a range of VR locomotion techniques and is an invaluable resource for VR developers. One key aspect of this is the concept of categories; this allows each technique to be rated against a set of criteria that is of design or accessibility relevance to the implementing developer. While such a resource does not exist, the categories below are proposed as suitable 'interaction vault' categories with this paper's technique being rated in each proposed category.

Accessibility

Does this technique rely on assumptions that exclude a portion of the population? Does it include fine motor control, visual perception (colour, granularity) or other aspects that may limit the ability for all users to employ it? Low accessibility requires full function over a range of senses and motor control. High accessibility does not introduce any additional barriers over regular VR interaction.

This technique rates as moderate to high accessibility. It does rely on visual perception of the offset so the effect may be negligible on users with inherent lack of visual perception and limb cognition however it will not usually undermine the core experience. The physical limb movement is more likely to be a factor. Where users have limited limb movement this technique will magnify difficulties in gross motor movement. The mitigating factor here is for users that have alternative interaction devices, the technique only relies on the control input location and can be readily adapted to other control schemes. Additionally, developers can open up the tuning parameter k as a user customizable parameter to increase/decrease the effect based on personal requirements.

Sustainability

How much effort does the technique take? Is this something a user is likely to be able to do for a sustained period of time or is it likely to take a physical toll? A highly sustainable technique is one that can be conducted almost indefinitely. A technique rating low in this category is one that users are unlikely to be able to continue over any period of time due to physical, mental or other impacts.

This technique rates as moderate. It is important to note that the underlying experience is the main factor (i.e.. How much lifting is involved); this technique does however increase the physical effort somewhat through increased range of movement requirements. This increase and the fact the core interaction is a gross motor

[1] https://locomotionvault.github.io/

movement means that it is not indefinitely sustainable. This is reinforced by one of the key findings of Moosavi et al (2024) that showed pseudo-haptics triggered the same biological responses as weightlifting.

Embodiment Level

How much does this interaction technique support embodiment over a broad range of applications? Does it require specific settings to maintain embodiment? **A technique rating high in this category is a natural extension of the user in the VR environment. A technique rating low on this scale is one that by its nature requires the user to separate themselves from their current experience.**

This technique is presented as a hand lifting an object however it is quite broadly applicable; the technique is "lifting element" agnostic and the C/D ratio applies equally to human limbs, robot arms and forklift tines. The technique as presented does not rely on additional visualisations or indications so rates highly in the embodiment category.

Groundedness

Is the technique grounded in real world interaction? This category considers how close to everyday life the interaction is and gives a sense of 'intuitiveness'. Note that high groundedness only indicates it is similar to real world interactions and is not objectively the best option; low groundedness may be preferable depending on the experience intent, especially where the experience is surreal or seeks to evoke magic or science fiction feelings.

This technique is highly grounded in reality. The technique imposes no additional mechanical or other requirements and is primarily effective due to brain interpretation of the C/D ratio offset.

Evidence based best practice

Has this technique been assessed in academic trials? A technique that rates high in this category is a widely implemented technique with clear lessons learned. A low rating in this category is a novel technique that has not been thoroughly battle tested.

This technique as presented is medium-high on the best practice scale. The underlying concepts are incredibly well supported over decades of research. Despite this, the VR implementation as a technique is less well documented. The implementation proposed in this paper is well grounded within its scope but does present edge cases and challenges for broader applications, discussed further in the Future Extensions section below.

Applicability

Is this technique suited to a narrow range of use cases or is it broadly applicable over a range of experiences? A technique rating highly in this category is suitable as

a general base for developers to extend or is general enough that it is experience agnostic. A technique rating low in this category requires exact and complete implementation as outlined to be effective and/or is very specific in its use case so will not be suitable for a range of experiences.

This technique is broadly applicable to a range of experiences. Any experience that requires object manipulation can enhance user presence and immersion through this technique.

Future Extensions

This paper has outlined an approach for emulating weight in VR without any additional equipment. This paper has been heavily research informed and as such presents the approach based on findings of the reviewed papers. It is noted that there are gaps and limitations with the proposed technique and future research, or testing is proposed to address the following limitations with the proposed approach:

- C/D ratio by spatial axis for full 3D weight emulation.
- Characterisation of muti modal pseudo-haptics including audio, VFX and controller shake.
- Multi hand implementation and object centre of gravity considerations including torque.
- Quantifying increased task time impacts due to C/D ratio.
- Investigation of impacts of habituation in purely virtual environments and sensitivity post C/D ratio removal.

Generative AI Acknowledgement

Generative AI was used to support the research and development of this report. Specifically:

- Elicit was used to conduct a broad literature review in parallel with manual literature review.
- Relevant sources from the Elicit review were manually reviewed to confirm the stated points were relevant or (in most cases) to extract points deemed specifically relevant. **NB: This is critical for Elicit in particular; while excellent at identifying and collating related academic work, many of the conclusions outlined in the summary were not reflected in the cited work.**
- The summary of all referenced material (personal summary of key parts, copied selections of areas of particular interest, personal thoughts, connections and links) were passed to Gemini with the prompt: "*Please review this and provide: 1. A summary of the key points, including any linkages not identified. 2. An overview of any links stated that are not fully supported with referenced material or any other gaps identified. 3. A summary mapped to the skeleton of the provided document.*"
- The output of this was used to review additional sources and iterate on the structure, linkages and supporting content of the document.
- The initial state was preliminary research and personal thoughts on the general technique – all focused on the "Technique Discussion" section. The Description and Categorisation areas were only addressed once the research informed discussion had developed a base for the proposed technique.
- At incremental stages, paper drafts were provided to Gemini to confirm cohesion, flow and any areas that were vague or unclear.
- No generative AI products were used directly; all outputs were used to support parallel manual work and/or to review claims and structure.

References

Burns, E., Razzaque, S., Panter, A., Whitton, M., McCallus, M., & Brooks, F. (2005). The Hand is Slower than the Eye: A quantitative exploration of visual dominance over proprioception . *IEEE Virtual Reality*.

Dominjon, L., Lecuyer, A., Burkhardt, J.-M., Richard, P., & Richir, S. (2005). Influence of control/display ratio on the perception of mass of manipulated objects in virtual environments. *IEEE Proceedings. VR 2005*.

Gibson, J. (1933). Adaptation, after-effect and contrast in the perception of curved lines. *Journal of Experimental Psychology, 16*(1).

Hartfill, J., Gabel, J., Kruse, L., Schmidt, S., Riebandt, K., Kühn, S., & Steinicke, F. (2021). Analysis of Detection Thresholds for Hand Redirection during Mid-Air Interactions in Virtual Reality. *27th ACM Symposium on Virtual Reality Software and Technology*.

Ito, K., Ban, Y., & Warisawa, S. (2024). Influence of habituation on pseudo-haptic weight perception of virtual objects. *Frontiers in Virtual Reality*.

Moosavi, M., Raimbaud, P., Guillet, C., & Mérienne, F. (2024). Enhancing weight perception in virtual reality: an analysis of kinematic features. *Virtual Reality*.

Palmerius, K., Johansson, D., Höst, G., & Schönborn, K. (2014). An Analysis of the Influence of a Pseudo-haptic Cue on the Haptic Perception of Weight. *Haptics: Neuroscience, Devices, Modeling, and Applications. EuroHaptics 2014.* (pp. 117–125). Berlin: Springer.

Samad, M., Gatti, E., Hermes, A., Benko, H., & Parise, C. (2019). Pseudo-Haptic Weight: Changing the Perceived Weight of Virtual Objects By Manipulating Control-Display Ratio. *Proceedings of CHI Conference on Human Factors in Computing Systems*.